Activity #1 – Variables

Instructions

- Carefully analyze the given Java program.
- Trace the references and memory changes step-by-step.
- Complete all required tables and questions.
- Show your logical reasoning where necessary.
- Write your answers in a yellow paper (handwritten).
- Take a clear photo or scan of your work.
- Upload your file in Canvas on or before the deadline.
- Submission Format: Section_LastName_FirstName_CodeAnalysisActivity#1.pdf

Please review the code below.

```java
class Student {
    String name;
    int grade;

    Student(String n, int g) {
        name = n;
        grade = g;
    }
}

public class TestStudent {
    public static void main(String[] args) {

        Student s1 = new Student("John", 85);
        Student s2 = new Student("Maria", 90);
        Student s3 = s2;

        s1.grade = 95;
        s3.name = "Ana";

        System.out.println(s1.name + " " + s1.grade);
        System.out.println(s2.name + " " + s2.grade);
        System.out.println(s3.name + " " + s3.grade);
    }
}
```

No. _____
Date _____

ACTIVITY # 1 - VARIABLES

CORTEZO, MARKY G.
BSIT 2-I

COMP 009 - Object-Oriented Programming
Date: March 3, 2026

PART A: Variable Identification Table

Variable Name	Data type	Variable Type	Stored In
name	string	Instance Variable	Heap
grade	int	Instance Variable	Heap
s1	class	Reference Variable	stack
s2	class	Reference Variable	Stack
s3	class	Reference Variable	Stack

PART B - Explanation

Statement	What happens?	Memory Impact
Student s1 = new Student ("John", 85);	A new student object created with name "John" and grade = 85. s1 then references the object.	An object is created in the heap, s1 in the stack stores its memory address.
Student s2 = new Student ("Maria", 90);	A second student object created with name "Maria" and grade = 90. s2 references the object.	A second subject created in the heap; s2 in the stack stores its memory address.
student s3 = s2	The reference held by s2 is assigned to s3.	s3 now points the same object as s2 in the heap; No new object is created.
s1.grade = 95;	The grade field of the object referenced by s1 is now updated.	The value 85 is now replaced by 95 in the first student object in the heap.
s3.name = "Ana";	The name field of the object referenced by is updated to "Ana".	Because s3 & s2 share the same object, the name, "Maria" is now changed to "Ana" for both.

PART C - OUTPUT:

John 95

Ana 90

Ana 90